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C++ generic overload functions

Experimental overload functions for C++17. This paper proposes two functions that allow to overload lambdas or function objects, but also member and non-member functions.

Contents

Introduction.....	1
Motivation and Scope.....	1
Why do we need an overload function?.....	2
Design rationale.....	3
Which kind of functions would overload accept.....	3
Binary or variadic interface.....	3
Passing parameters by value or by forward reference.....	3
reference_wrapper<F> to deduce F&	4
Selecting the best or the first overload.....	4
Result type of resulting function objects.....	4
Result type of overload/first_overload.....	4
Open points.....	5
Technical Specification.....	5
Header <experimental/functional> Synopsis.....	5
Implementation.....	7
Acknowledgements.....	7
References.....	7

Introduction

This paper proposes two functions that allow to overload lambdas or function objects, but also member and non-member functions:

- `overload` selects the best overload using C++ overload resolution and
- `first_overload` selects the first overload using C++ overload resolution.

Motivation and Scope

As lambdas functions, function objects, can't be overloaded in the usual implicit way, but they can be "explicitly overloaded" using the proposed `overload` function:

This function would be especially useful for creating visitors, e.g. for variant.

```

auto visitor = overload(
    [](int i, int j )           { ... },
    [](int i, string const &j ) { ... },
    [](auto const &i, auto const &j ) { ... }
);

visitor( 1, std::string{"2"} ); // ok - calls (int,std::string) "overload"

```

The overload function when there are only two parameters could be defined as follows (this is valid only for lambdas and function objects)

```

template<class F1, class F2> struct overloaded : F1, F2
{
    overloaded(F1 x1, F2 x2) : F1(x1), F2(x2) {}
    using F1::operator();
    using F2::operator();
};

template<class F1, class F2>
overloaded<F1, F2> overload(F1 f1, F2 f2)
{ return overloaded<F1, F2>(f1, f2); }

```

Why do we need an overload function?

Instead of the previous example

```

auto visitor = overload(
    [](int i, int j )           { ... },
    [](int i, string const &j ) { ... },
    [](auto const &i, auto const &j ) { ... }
);

```

the user can define a function object

```

struct
{
    auto operator()(int i, int j ) { ... }
    auto operator()(int i, string const &j ) { ... }
    template <class T1, class T2>
    auto operator()(T1 const &i, T2 const &j ) { ... }
} visitor;

```

So, what are the advantages and liabilities of the overload function.

First the advantages:

1. With overload the user can use existing functions that it can combine, using the function object would need to define an overload and forward to the existing function.
2. The user can group the overloaded functions as close as possible where they are used and don't need to define a class elsewhere. This is in line with the philosophy of lambda functions.
3. Each overload can have its own captured data, either using lambdas or other existing function objects.
4. Any additional feature of lambda functions, automatic friendship, access to this, and so

forth.

5. `first_overload` has the additional advantage that it can not be achieved with a function object as it changes the resolution rules.

Next the liabilities:

1. The overload function generates a function object that is a little bit more complex and so would take more time to compile.
2. The the result type of overload function is unspecified and so storing it in an structure is more difficult (as it is the case for `std::bind`).
3. With the function object the user is able to share the same data for all the overloads.

Note that that the last point could be seen as an advantage and a liability depending on the user needs.

Design rationale

Which kind of functions would overload accept

The previous definition of `overload` is quite simple, however it doesn't accept member functions nor non-member function, as `std::bind` does, but only function objects and lambda captures.

As there is no major problem implementing it and that their inclusion doesn't degrade the run-time performances, we opt to allow them also. The alternative would be to force the user to use `std::bind` or wrap them with a lambda.

Binary or variadic interface

We could either provide a binary or a variadic `overload` function.

```
auto visitor =
overload([](int i, int j )           { ... },
overload([](int i, string const &j ){ ... },
[] (auto const &i, auto const &j ) { ... }
));
```

The binary function needs to repeat the `overload` word for each new overloaded function.

We think that the variadic version is not much more complex to implement and makes user code simpler.

Passing parameters by value or by forward reference

The function `overload` must store the passed parameters. If the interface is by value, the user will be forced to `move` movable but non-copyable function objects. Using forward references has not this inconvenient, and the implementation can optimize when the function object is copyable.

This has the inconvenient that the move is implicit. We follow here the same design than `when_all` and `when_any`.

`reference_wrapper<F>` to deduce `F&`

As with other functions that need to copy the parameters (as `std::bind`, `std::thread`, ...), the user can use `std::ref` to pass by reference.

The user could prefer to pass by reference if the function object is state-full or if the function object is expensive to move (copy if not movable) or even s/he would need it if the function object is not movable at all.

Selecting the best or the first overload

Call the functions based on C++ overload resolution, which tries to find the best match, is a good generalization of overloading to lambdas and function objects.

However, when trying to do overloading involving something more generic, it can lead to ambiguities. So the need for a function that will pick the first function that is callable. This allows ordering the functions based on which one is more specific.

As both cases are useful, the proposal includes two functions: `overload` and `first_overload`.

- `overload` selects the best overload using C++ overload resolution and
- `first_overload` selects the first overload using C++ overload resolution.

Fit library name `match` and `conditional` respectively. FTL uses `match` to mean `first_overload`.

Result type of resulting function objects

The proposed `overload` and `first_overload` functions don't add any constraint on the result type of the overloaded functions. The result type when calling the resulting function object would be the one of the selected overloaded function.

However the user can force the result type and in this case the result type of all the overloads must be convertible to this type (contribution from Matt Calabrese).

This can be useful in order to improve the compile time of a possible match/visit function that could take advantage when it knows the result type of all the overloads.

Result type of `overload/first_overload`

The result type of these functions is unspecified as it is the result type of `std::bind`.

However when the functions objects have a state it will be useful that the user can inspect the state. The result type should provide an overload for `std::get<F>/std::get<I>` functions (contribution from Matt Calabrese).

These functions should take in account that the overload can be a `reference_wrapper<F>` in order to allow `get<F&>(ovl)`.

Open points

The authors would like to have an answer to the following points if there is at all an interest in this proposal:

- **Should the callable be passed by value, forcing the use of `std::move`?**
- **A better name for the proposed functions?**
best_overload/first_overload, overloads/cases.
- **Do we want to expose the result type of these functions?**
- **Do we want to inspect the state of the result type of these functions via a `get<T>/get<I>`?**

Technical Specification

Note: The following wording needs of course much more details to describe the behavior of the unspecified result type using `DECAY_COPY` and `INVOKE`.

The wording doesn't contains yet the `get<T>/get<I>` tuple like functions.

Header `<experimental/functional>` Synopsis

Add the following declaration in `experimental/functional`.

```
namespace std {
namespace experimental {
inline namespace fundamental_v2 {
    template <class ... Fs>
        'see below' overload(Fs &&... fcts);
    template <class R, class ... Fs>
        'see below' overload(Fs &&... fcts);
    template <class ... Fs>
        'see below' first_overload(Fs &&... fcts);
    template <class R, class ... Fs>
        'see below' first_overload(Fs &&... fcts);
}
}
}
```

Function Template `overload`

```
template <class R, class ... Fs>
'see below' overload(Fs &&... fcts);
```

Requires: Fs are Callable and Movable and the result type of each parameter must be convertible to

R.

Result type: A function object that behaves as if all the parameters were overloaded when calling it. The result type will contain the nested `result_type` type alias R. The call to an instance of this type will select the best overload. If there is not such a best overload, either because there is no candidate or that there are ambiguous candidates, the invocation expression will be ill-formed.

Returns: An instance of the result type, that contains a decay copy of each one of the arguments.

Throws: Any exception thrown during the construction of the resulting function object.

```
template <class ... Fs>
    'see below' overload(Fs &&... fcts);
```

Requires: Fs are Callable and Movable.

Result type: A function object that behaves as if all the parameters were overloaded when calling it. The call to an instance of this type will select the best overload matching the invocation. If there is not such a overload the invocation expression will be ill-formed.

Returns: An instance of the result type, that contains a decay copy of each one of the arguments.

Throws: Any exception thrown during the construction of the resulting function object.

Function Template `first_overload`

```
template <class R, class ... Fs>
    'see below' first_overload(Fs &&... fcts);
```

Requires: Fs are Callable and Movable and the result type of each parameter must be convertible to R.

Result type: A function object that behaves as if all the parameters were overloaded when calling it. The result type will contain the nested `result_type` type alias R. The call to an instance of this type will select the first overload matching the invocation. If there is not such a overload the invocation expression will be ill-formed.

Returns: An instance of the result type, that contains a decay copy of each one of the arguments.

Throws: Any exception thrown during the construction of the resulting function object.

```
template <class ... Fs>
    'see below' first_overload(Fs &&... fcts);
```

Requires: Fs are Callable and Movable.

Result type: A function object that behaves as if all the parameters were overloaded when calling it. The call to an instance of this type will select the first overload matching the invocation. If there is not such a overload the invocation expression will be ill-formed.

Returns: An instance of the result type, that contains a decay copy of each one of the arguments.

Throws: Any exception thrown during the construction of the resulting function object.

Implementation

There is an implementation of `overload` at <https://github.com/viboes/tags>.

Acknowledgements

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Thanks to Fit and FTL authors from where the idea of the `first_overload` function comes from.

Thanks to Matt Calabrese for its useful suggestions.

References

- Fit : <https://github.com/pfultz2/Fit>
- FTL: <https://github.com/beark/ftl>